PRECISION IMU CARRIER

ARK ADIS16507

Precision 6-DoF MEMS IMU carrier based on the ADIS16507-2 or ADIS16507-3

The USA-built ARK ADIS16507 is a compact carrier for the Analog Devices ADIS16507 precision MEMS IMU, sold with the ADIS16507-2 or ADIS16507-3. It provides a factory-calibrated triaxial gyroscope (±500 °/s or ±2000 °/s) and triaxial accelerometer (±40 g) over SPI, with onboard 3.3 V regulation from a 5 V supply.

The 11-pin JST-GH connector matches the ARK PAB SPI pinout for direct use with ARK flight controllers and carriers.



KEY FEATURES

- ADIS16507-2 or ADIS16507-3 precision MEMS IMU
- Gyro: ±500 °/s (-2) or ±2000 °/s (-3)
- Accel: ±40 g (±392 m/s²)
- Factory calibrated sensitivity, bias, and alignment
- SPI interface with data-ready
- Onboard 3.3 V LDO from 5 V input
- 11-pin JST-GH matching ARK PAB SPI pinout
- Includes cable

SPECIFICATIONS

ELECTRICAL	
Supply voltage	5 V
Supply current	~41 mA average
IMU supply	3.3 V (onboard LDO)
Interface	SPI

ENVIRONMENTAL	
Operating temperature	−25 °C to +85 °C
Storage temperature	−25 °C to +85 °C
IMU operating temperature	−40 °C to +105 °C
IMU storage temperature	−65 °C to +150 °C
IMU mechanical shock	2000 g, 0.5 ms ½ sine, any axis, unpowered

MECHANICAL	
Dimensions	22.5 × 25.0 × 7.1 mm
Weight	4 g

AT A GLANCE	
IMU	ADIS16507-2 / -3
Gyro	±500 / ±2000 °/s
Accel	±40 g
Interface	SPI
Supply	5 V
Dimensions	22.5 × 25 × 7.1 mm
Weight	4 g
Hardware rev	Rev 1

COMPLIANCE & SOURCING

› Designed and manufactured in the USA

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-adis16507-2/
arkelectron.com/product/ark-adis16507/

DOCS docs.arkelectron.com/products/imu/ark-adis16507

CAD github.com/ARK-Electronics/ark-hardware

VARIANTS

	ARK ADIS16507-2	ARK ADIS16507-3
SKU	ARK-ADIS16507-2	ARK-ADIS16507-3
Gyro range	±500 °/s	±2000 °/s
Gyro in-run bias stability	2.7 °/h	7.5 / 8.1 / 4.9 °/h (X / Y / Z)

Same board, connector and pinout. Both: accelerometer ±40 g, in-run bias stability 12.7 µg (X, Y) and 13.7 µg (Z). Bias stability is the ADIS16507 data sheet typical.

CONNECTOR PINOUT

PIN	SIGNAL	NOTES
1	VDD_5V	5 V supply
2	SPI_SCK	
3	SPI_MISO	
4	SPI_MOSI	
5	SPI_nCS	
6	NC	
7	SPI_nSYNC	
8	SPI_DRDY	Data ready
9	NC	
10	SPI_nRESET	Reset, 10 kΩ pull-up to 3.3 V
11	GND	

Connector: 11-pin JST-GH (SM11B-GHS-TB). Matches the ARK PAB SPI pinout (nCS2 / DRDY2 unused). Signals are 3.3 V.

NOTES

- Factory calibrated over −40 °C to +85 °C. Extended exposure outside −40 °C to +105 °C can degrade calibration accuracy.
- IMU environmental ratings are the ADIS16507 absolute maximum ratings. No vibration survival limit is specified.
- SPI-compatible data communications with programmable operation and control.
- Includes 11-pin JST-GH cable.

ARK-DS-ARK-ADIS16507 | Version 1.0, 2026-10-29 | Compiled from the ARK ADIS16507 Carrier Production schematic Rev 1, the Production bill of materials, manufacturing documentation, and ARK Electronics published documentation. IMU dynamic ranges per ADIS16507-2BMLZ and ADIS16507-3BMLZ; IMU environmental ratings per the Analog Devices ADIS16507 data sheet Rev. A. Board temperature rating per component data sheets; the JST-GH connector sets −25 °C. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.